NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
B.TECH 5TH SEM IT, AUTOMATA THEORY (ITPC37)
SESSIONAL-II

Duration: - 50 Minutes.

M.M 15

Note: - All questions are compulsory.

Q-1) Prove following grammar is ambiguous and convert the same into CNF.

$S \rightarrow iCtS \quad S \rightarrow iCtSeS \quad S \rightarrow a \quad C \rightarrow b$

(3)

Q-2) Construct a PDA for the following $L = \{a^n c b^{2n} / n \geq 1\}$ over alphabet {a,b,c} and show its working for the string aacbbbb.

(3)

Q-3) Define DPDA. Give DPDA for following langauge.

$L = \{ w \in (a,b)^* \mid n_0(w) > n_1(w) \}$

(3)

Q-4) Give PDA for the following :

$L = \{ w \in (a,b)^* \mid n_0(w) = n_1(w) \}$

Also show its working for the string abaabab.

(3)

Q-5) Give regular expression for each of the following:

(3)

i. $\{w \varepsilon \{a,b\}^* \mid$ bb is not a substring of w $\}$.

ii. $\{w \varepsilon \{0,1\}^* \mid$ w ends with 0 and does not contain the substring 11.

iii. $\{w \varepsilon \{0,1\}^* \mid$ w contain at least one of the two substrings 10 or 001$\}$.